

Look-Ahead Bias Audit — One-Page Sign-Off

Project: Backtesting the ProntoNLP Earnings-Call ATC Signal · **Author:** Yueqi Lin · **Reference:** Student Handout §3

#	Item	Status & Evidence (notebook cell / test ID)
3.1	Entry timing (AMC vs BMO)	PASS — <code>hour >= 16 UTC</code> -> next NYSE day; <code>hour < 16 UTC</code> -> same day. Assertion in <code>00_data_prep.ipynb</code> cell 19. Gray-zone (13–15 UTC) rule documented in cell 18.
3.2	Forward returns are targets only	PASS — <code>{return_1d ... return_20d}</code> disjoint from <code>feature_cols</code> (assertion in cell 25, under the §9b markdown header at cell 24). T2: 0 overlaps / 772 features.
3.3	Cross-sectional features are PIT	PASS — All Z-scores / sector ranks deferred to <code>01_analysis.ipynb</code> walk-forward; per-fold on training events only.
3.4	Feature selection inside training	PASS — IC top-30 sparse selection re-fit on each fold’s training data; no full-sample selection. T15 (<code>07_audit_gap_tests.py</code>) asserts selection is data-dependent (Jaccard 0.05 across disjoint windows).
3.5	Imputation / scaling inside training	PASS — <code>StandardScaler.fit(train)</code> then <code>transform(test)</code> inside walk-forward loop.
3.6	Universe membership is PIT	PASS — RU3K fully PIT via CRSP top-3000 mcap (annual June reconstitution; T11 = 0 violations / 153,988 events). SP500/SP1500 current-composition caveat documented per handout §6.3.
3.7	INGESTDATEUTC is not availability date	PASS — Inspected: 1,658d mean lag (batch backfill). Entry date uses <code>MOSTIMPORTANTDATEUTC</code> only; rationale in cell 18. T6: 0 INGEST features in model input.
3.8	No “future” QoQ deltas	PASS — All QoQ/2Q/YoY deltas use <code>groupby(BESTTICKER).shift(1/2/4)</code> on data sorted by <code>entry_date</code> ascending. T8 verified.
3.9	Corporate-action / delisting	PASS — NaN-return events excluded from all analyses (no roll-forward, no imputation). Documented in <code>00_data_prep.ipynb</code> §9b and research PDF §7. T16 (<code>07_audit_gap_tests.py</code>) asserts no mass <code>fillna(0)</code> in <code>return_20d</code> .
3.10	Hyperparameter tuning	PASS — Tuning sub-period 2010–2017; walk-forward test 2018Q1+ uses frozen hyperparameters. No full-sample grid search. T17 (<code>07_audit_gap_tests.py</code>) asserts no <code>GridSearchCV</code> / <code>Optuna</code> / <code>hyperopt</code> in <code>01_analysis.ipynb</code> .

Programmatic verification: T1–T8 in `02_lookahead_tests.ipynb`, T9–T14 in `06_wrds_lookahead_tests.py`, T15–T17 in `07_audit_gap_tests.py` — **all 17 tests PASS**.

Detailed implementation notes for each item are preserved in `reports/look_ahead_audit_detail.md`.

I certify that all ten look-ahead audit items above have been reviewed and pass. The backtest implementation in `00_data_prep.ipynb` and `01_analysis.ipynb` contains no look-ahead bias that I am aware of.

Signed: Yueqi Lin · **Date:** May 2026 · **Course:** LLM-Driven Quant Research — Final Assignment